Ecological engineers

Prairie dogs in the American Midwest are a good example of a keystone species whose impact is the result of their “engineering” activities. Huge colonies of these small mammals dig networks of tunnels beneath the prairie grasslands. They sleep and raise their young in these extensive burrows, converting the grassland into a suitable habitat.

The prairie dogs’ constant digging dramatically increases soil turnover and allows nutrients and water from rain and snow to penetrate deeper than would otherwise be the case. The damp, nutrient-rich soil encourages a diversity of plants, and birds such as Mountain Plovers feed and nest in the short grass. Predators like Ferruginous Hawks and black-footed ferrets are attracted to the area to hunt for prey, and the ferrets and tiger salamanders use the burrows for shelter. Almost 150 species of plant and animal are known to benefit from prairie dog colonies. Although there are “losers”—notably vertebrates that favor tall vegetation—the prairie dogs’ presence increases overall biodiversity. When colonies die out, scrubby patches of mesquite vegetation replace short grasses, plovers abandon the area, and predator numbers decline.